

COMPETITION RULES

IEEE VICTORIAN
SECTION
INTER-UNIVERSITY
ROBOTICS CHALLENGE
2026

Objective

Your objective is to build a robot that can follow a line and retrieve an object to the start line on its own, quickly and accurately. The track will have a black line on a white surface, and your robot's job is to stay on the line, retrieve the object, and finish the course as fast as possible.

Track Specifications

There will be two tracks; a practice track for participants to test their robots, and a final track which will be revealed on competition day. The specifications of the tracks are given as follows:

- The field is constructed out of a white material with dimensions 2400x1200 (final track), 1200x1800 (practice track)
- The track is open, except for the object at the end of the track.
- A track is laid on the field using black PVC electrical tape.
- The thickness of the track ranges from 9mm to 18mm.
- The track includes checkpoints to monitor and record the progress of each participant.
- The checkpoints are marked by colored pin (green, yellow, red, blue)
- A section is the region between two checkpoints.
- The start and end of the track are marked by a black box. The start has a smaller black box than the end.
- The track layout may vary and can include intersections, turns, curves, gaps, and multiple possible routes to reach the end of the track.
- The object to collect is a ping-pong/table-tennis ball.
- The object will be placed in the middle of the black box at the end of the track.

Any form of rude or disruptive behaviour may result in disqualification.

Organisers reserve the right to change the rules at any time.

The Robot

- The robot must be autonomous.
- To ensure fair competition, the robot must be built entirely with the components provided by the organisers and the use of a personal component is only permitted if it is the same model and specification.
- The frame of the robot must cover the track at all times.

Competition Procedure

- Each team will have one 7-minute trial run on the final track, followed by a single 7-minute competition attempt.
- The time taken in each section will be recorded.
- Progress will be measured by checkpoints, counted when the robot's IR sensors cross the checkpoint line.
 - The robot must start fully inside the starting box.
- If the robot is adjusted by hand, a time penalty will be applied: 5 seconds for the first touch, 10 seconds for the second, and 15 seconds for each subsequent touch.
- Each team may skip only one section during their competition attempt. It must be declared before entering that section. The end section may not be skipped. Skipping when the robot has the object is not allowed.
- Skipping to a green checkpoint incurs a 20-second penalty, yellow a 40-second penalty, and red a 60-second penalty.
- Skipping to the blue checkpoint, which marks the start and finish, is not allowed.
- The robot must always be traversing over the black tape and is only permitted to be outside of the black tape when returning to a line to traverse back to the start.
- The robot may only begin looking for the object when it enters the end box. The robot must not let the object exit the end black box. If the object exits the black box, the participants are allowed to readjust the object back to the centre for a 20-second penalty.
- Once the robot has the object, it must traverse backwards to the start black box with the object through the track it travelled to get to the end black box.
- The robot's full body and object must stop within the start box to complete the track.
- The aim is to reach the farthest checkpoint in the shortest adjusted time, with ties decided by the lowest adjusted time.
- An attempt ends when the robot stops inside the start black box with the object, or when the 7-minute limit expires, or the team decides to end the attempt.
- The referees have the final say in any situation.

FAQs

Q1. Can I break my breadboard in half

- **Sure! We respect your design choice**

Q2. Does the robot have to “pick up” the ball?

- **Nope, as long as your robot and ball can reach the start from end together, the method is up to your choices! You can choose to let robot pushes the ball... etc.**

Q3. Can I use rubber band/cable tie to tidy up wires

- **Yes, only for wire managements**